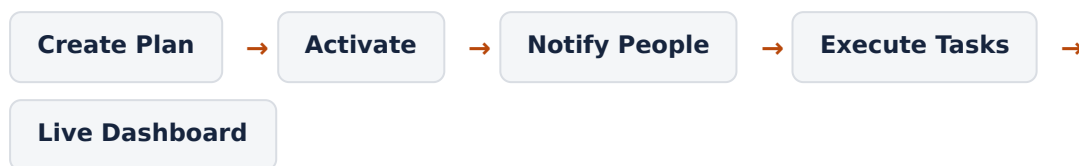


FEATURES & REQUIREMENTS SPECIFICATION

EXECPLAN

Operational Plan Activation & Live Execution Tracking

نظام إكسك بلان — تفعيل الخطط ومتابعة التنفيذ لحظة بلحظة



Document: Product Features & Requirements Report · **Version:** MVP 1.0 · **Date:** June 2026
Status: All four build phases complete and verified · **Region:** Kuwait / GCC · **Language:** Arabic-first (RTL), English locale included

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1 · Introduction & Scope

This document specifies the complete set of features and requirements for **EXECPLAN**, an operational plan activation and live execution tracking system built for shift-based organizations in Kuwait and the wider GCC. It is the consolidated reference for what the system does, the rules it enforces, and the conditions under which a feature is considered delivered.

EXECPLAN is deliberately **not** a general project-management tool. Its entire design serves one operational cycle: a plan manager prepares a reusable plan, activates it in one action, the right people on the current shift are notified and acknowledge their readiness, tasks are executed and checked off, and the whole organization watches progress on a live dashboard. Non-responders are escalated automatically until their designated substitutes take over.

The scope of this report is the delivered **MVP 1.0**: a Django REST backend, an Arabic-first Next.js web application, and an Expo (React Native) mobile application, sharing one data model and one API. Items consciously excluded from the MVP are listed in Section 13 as the forward roadmap.

2 · Product Vision & Core Principles

The product answers the three questions an operations chief asks during any activation: *who has actually mobilized, which teams are ahead or behind, and what is overdue*. Every feature exists to answer one of these faster and more reliably than phone trees and messaging groups can.

Guiding principles (binding on all features)

- **One cycle, not a toolkit.** Features map onto Create → Activate → Notify → Execute → Dashboard. Anything outside the cycle is out of scope for the MVP.
- **Template versus snapshot.** A plan is a reusable template; activating it freezes an immutable snapshot. Editing a plan never disturbs a running activation.
- **One definition of "responded."** A response is the participant tapping the readiness button «أنا جاهز» in the app. Nothing else counts.
- **Built for non-technical field staff.** Big buttons, one primary action per screen, minimal text, clear Arabic labels, no jargon.
- **Arabic-first.** Right-to-left throughout, Kuwait timezone and shift conventions native; a complete English locale ships alongside.
- **Operationally lean.** Runs on a single modest server: polling instead of WebSockets, no background-job infrastructure required for the MVP.

3 · User Roles & Personas

Four roles are recognized. Each role determines the home screen, the available actions, and the slice of data the user can see.

Role	Platform	Responsibilities & what they can do
System Admin مدير النظام	Web	Manage organizations, departments, and users; assign roles; full visibility across the system.
Plan Manager مدير الخطة	Web + Mobile	Create and edit plans, define teams, tasks and shift rosters, activate plans, watch the live dashboard, broadcast messages, run escalation, close activations.
Team Leader قائد الفريق	Mobile-first	See his team, assign plan tasks to members, set a substitute per member, monitor team progress, raise issues to the plan manager.
Team Member عضو الفريق	Mobile	Receive activation notification, acknowledge readiness, see only his own tasks, check tasks done, add a note, read broadcasts.

4 · Functional Features — by Module

The delivered functionality grouped into seven modules. Each line is a shipped capability of MVP 1.0.

4.1 · Authentication & Identity

- Username/password sign-in issuing JWT access and refresh tokens.
- Automatic, transparent access-token refresh; expired sessions return the user to login.
- Persistent session on mobile (secure on-device storage) and web (browser storage).
- Role-based routing to the correct home screen immediately after login.

4.2 · Organization & User Administration

- Create and list organizations and departments.
- Create users with role, name, phone, and department; tabular listing of all users.
- Role assignment from the four-role set; admin-only for all write operations.

4.3 · Plan Authoring

- Create a plan with name, type (nine categories: daily, weekly, emergency, guard, transport, maintenance, IT, inspection, general), objective, description, and scope.
- Important contacts and emergency numbers stored as structured lists.
- Define teams within a plan, each with an optional team leader.
- Assign members to teams.
- Define task templates per team, each with a title, order, and duration.
- Build a shift roster: assign people to a team for a shift and date, with an optional substitute per person.
- Four-step guided creation wizard with validation at each step.
- Designate which users are authorized to activate the plan.

4.4 · Activation

- One-action activation behind a plain-language confirmation.
- Immutable snapshot: participants resolved from the current shift roster, tasks generated from templates, team names frozen as text.
- Per-participant in-app notification and first call attempt recorded at activation.
- Guards: refuses to activate if a plan is already running or if no one is on duty for the current shift; activation restricted to authorized users.

4.5 · Execution & Response

- Readiness acknowledgment « أنا جاهز» — the sole response mechanism.

- Per-member task checklist: mark done, undo, add and save a note per task.
- Members see strictly their own tasks; never anyone else's.
- Team leaders reassign tasks among their own team's members.
- Team leaders set or change a participant's substitute on a live activation.
- Team leaders raise an issue to the plan manager, surfaced in the dashboard event log.

4.6 · Escalation

- Manual escalation cycle, triggered from the dashboard or via a management command.
- Each cycle adds a call attempt for every non-responder.
- At a configurable threshold (default five attempts) a non-responder is marked escalated and his frozen substitute is inducted automatically — with his own tasks, notification, and first call attempt.

4.7 · Live Monitoring, Broadcast & Closure

- Single aggregated live dashboard, auto-refreshing every 10 seconds.
- Counters: required, responded, not-responded, escalated, substitutes called.
- Response-rate and task-completion-rate indicators.
- Per-team ranking sorted best-to-delayed, with "best" and "delayed" badges.
- Overdue-tasks list and a merged chronological event log (latest 50 events).
- Broadcast a message to all participants (web and mobile); members read it in an in-app inbox.
- Close an activation and view a final summary of counts; live polling stops on closure.

5 · Screen Inventory

Web application (Next.js, Arabic RTL)

Screen	Role	Purpose
Login	All	Single-action sign-in.
Users administration	Admin	User table + add-user form.
Departments administration	Admin	Department table + add form.
My Plans خططي	Manager	Plan list with status badges.
Create-Plan Wizard	Manager	4 steps: معلومات الخطة · الفرق والأعضاء · المهام · النوبات والمراجعة.
Plan Detail	Manager	Big «إطلاق الخطة» button with confirmation.
Live Dashboard	Manager / Leader	Counters, rates, team ranking, overdue, event log, escalation & broadcast actions, close.
Activation Summary	Manager	Final counts after closure (no polling).

Mobile application (Expo, Arabic RTL)

Screen	Role	Purpose
Login	All	Sign-in.
Member Home	Member	Active-plan card, «أنا جاهز»  , task checklist with notes, broadcast inbox.
Leader Home	Leader	Team progress, task reassignment, substitute management, raise-issue.
Manager Home	Manager	Mini live dashboard, escalation, broadcast, close, plans with per-plan activate.

6 · Detailed Functional Requirements

Requirements use MoSCoW priority: **MUST** **SHOULD** **COULD** . All **MUST** items are delivered in MVP 1.0.

6.1 · Authentication

ID	Priority	Requirement
FR-AUTH-1	MUST	The system shall authenticate users by username and password and issue JWT access and refresh tokens.
FR-AUTH-2	MUST	The system shall refresh an expired access token automatically using the refresh token without forcing re-login.
FR-AUTH-3	MUST	On invalid or expired credentials the system shall return the user to the login screen.
FR-AUTH-4	MUST	The system shall route each user to a role-appropriate home screen after sign-in.

6.2 · Administration

ID	Priority	Requirement
FR-ADM-1	MUST	An admin shall create and list organizations and departments.
FR-ADM-2	MUST	An admin shall create users with a role, name, phone, and department, and view all users in a table.
FR-ADM-3	MUST	All user/department/organization write operations shall be restricted to the admin role.
FR-ADM-4	SHOULD	A plan manager shall be able to read the user list (read-only) to populate team membership during plan authoring.

6.3 · Plan Authoring

ID	Priority	Requirement
FR-PLAN-1	MUST	A manager shall create a plan with name, type, objective, description, and scope.
FR-PLAN-2	MUST	The plan shall store important contacts and emergency numbers as structured name/number lists.
FR-PLAN-3	MUST	A manager shall define teams within a plan and assign members to each team.

FR-PLAN-4	MUST	A manager shall define task templates per team, each with a title, ordering, and duration.
FR-PLAN-5	MUST	A manager shall build a shift roster assigning users to a team for a given shift and date, optionally designating a substitute per person.
FR-PLAN-6	MUST	The creation wizard shall present exactly four steps and validate each before advancing.
FR-PLAN-7	MUST	A manager shall designate which users are authorized to activate the plan.
FR-PLAN-8	MUST	Editing a plan shall never alter any existing activation derived from it.

6.4 · Activation

ID	Priority	Requirement
FR-ACT-1	MUST	Activation shall create an immutable snapshot recording participants, tasks, and team names at that instant.
FR-ACT-2	MUST	Activation shall resolve participants from the shift roster for the current shift and date.
FR-ACT-3	MUST	Activation shall generate one execution task per participant per applicable task template, with a due time derived from the template duration.
FR-ACT-4	MUST	Activation shall record one in-app notification and one initial call attempt per participant.
FR-ACT-5	MUST	Activation shall be rejected if the plan already has an active run, or if no one is on duty for the current shift.
FR-ACT-6	MUST	Activation shall be permitted only to the plan creator, its authorized activators, or an admin.

6.5 · Execution & Response

ID	Priority	Requirement
FR-EXE-1	MUST	A member shall acknowledge readiness with a single action; this is the only event counted as a response.
FR-EXE-2	MUST	A member shall mark his tasks done or not-done and add a note per task.
FR-EXE-3	MUST	A member shall see only his own tasks and only activations he participates in.
FR-EXE-4	MUST	A team leader shall reassign a task only to a member of a team he leads.
FR-EXE-5	MUST	A team leader shall set or change a participant's substitute on a live activation.

FR-EXE-6	SHOULD	A team leader shall raise an issue to the plan manager, recorded and shown in the dashboard event log.
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6.6 · Escalation

ID	Priority	Requirement
FR-ESC-1	MUST	An escalation cycle shall be triggerable from the dashboard and from a server management command, with identical behavior.
FR-ESC-2	MUST	Each cycle shall add one call attempt for every participant still pending.
FR-ESC-3	MUST	On reaching the configured attempt threshold without a response, a participant shall be marked escalated and his substitute inducted with tasks, notification, and a first call attempt.
FR-ESC-4	MUST	The escalation threshold shall be a configurable setting (default five).

6.7 · Monitoring, Broadcast & Closure

ID	Priority	Requirement
FR-MON-1	MUST	A single dashboard endpoint shall return all monitoring data in one response, polled every 10 seconds.
FR-MON-2	MUST	The dashboard shall present the five participant counters, response and task-completion rates, per-team ranking, overdue tasks, and the latest 50 events.
FR-MON-3	MUST	Per-team rows shall be sorted best-to-delayed to yield "best team" and "delayed team" indicators.
FR-MON-4	MUST	A manager shall broadcast a message to all participants; members shall read broadcasts in-app.
FR-MON-5	MUST	A manager shall close an activation; the system shall present a final counts summary and stop live polling.

7 · Business Rules & Logic

Rule	Definition
Shifts	Three fixed shifts in Asia/Kuwait: Morning 06:00–14:00, Evening 14:00–22:00, Night 22:00–06:00.
Night roster date	A night shift in progress after midnight resolves to the previous calendar day's roster, since the 22:00–06:00 shift belongs to the day it began.
On-duty resolution	At activation, for each required team the system reads who is assigned to the current shift today; that person is on duty and his substitute chain is frozen onto his participant record.
Shift-boundary	If an activation crosses a shift boundary, participants remain those frozen at activation time. Automatic handover is a future feature.
Response	A response equals the readiness tap. Opening the app, viewing tasks, or completing tasks do not count as a response.
Escalation	Non-responder → one call attempt per cycle → at threshold (default 5) marked escalated → substitute inducted with full task set, notification, and first call attempt.
Substitute source	The substitute is resolved from the roster's substitute link and frozen at activation; a leader may override it on a live activation, affecting subsequent cycles.
Notifications/calls (MVP)	Recorded as database entries only; no external delivery. A provider interface isolates future SMS/voice/WhatsApp integration.

8 • Data Model & Entities

Fifteen entities implement the MVP, divided between the template side and the runtime snapshot side.

Entity	Role in the system
User	Custom account with role, phone, organization, department.
Organization / Department	Organizational hierarchy.
Plan	Reusable template: metadata, contacts, authorized activators, status.
Team / TeamMembership	Teams within a plan and their member assignments.
ShiftAssignment	Roster row (user, team, shift, date, substitute-for) — answers "who is on duty".
TaskTemplate	Task blueprint per team (title, order, duration).
PlanActivation	Immutable runtime snapshot (status, shift, timestamps).
ActivationParticipant	A person frozen into an activation, with status and resolved substitute.
ExecutionTask	A task instance owned by a participant (status, note, due time).
NotificationLog	Record of an in-app notification or broadcast to a recipient.
CallAttempt	Numbered call-attempt placeholder per participant.
ResponseStatus	Marks that a participant acknowledged readiness, with timestamp.
EscalationLog	Record of an escalation and the substitute it produced.
BroadcastMessage	A manager's message to all participants of an activation.

9 · API Requirements

All endpoints are versioned under `/api/v1/` and secured by JWT. List endpoints are filtered by the requester's role.

Endpoint	Purpose
POST <code>/auth/login</code> · <code>/auth/refresh</code>	Obtain and refresh tokens.
CRUD: <code>organizations</code> , <code>departments</code> , <code>users</code> , <code>plans</code> , <code>teams</code> , <code>team-members</code> , <code>shift-assignments</code> , <code>task-templates</code>	Core data management.
POST <code>/plans/{id}/activate</code>	Create an activation snapshot.
GET <code>/activations/{id}/dashboard</code>	Single aggregated monitoring payload.
POST <code>/activations/{id}/acknowledge</code>	Record a member's readiness.
POST <code>/activations/{id}/run-escalation</code>	Run one escalation cycle.
PATCH <code>/execution-tasks/{id}</code>	Mark done, add note, or reassign.
POST <code>/activations/{id}/broadcast</code>	Message all participants.
POST <code>/activations/{id}/set-substitute</code>	Set/change a participant's substitute live.
POST <code>/activations/{id}/raise-issue</code>	Leader reports an issue to the manager.
GET <code>/activations/{id}/my-tasks</code> · <code>/my-notifications</code>	A member's own tasks and inbox.
POST <code>/activations/{id}/close</code>	Close the activation.

10 · Non-Functional Requirements

ID	Requirement
NFR-1	Real-time freshness. Dashboards and member views shall reflect a change within one 10-second polling interval.
NFR-2	Operational simplicity. The system shall run on a single internet-connected server with no WebSocket or background-job infrastructure in the MVP.
NFR-3	Localization. The interface shall be Arabic-first RTL; all user-facing strings shall live in translation files with a complete English locale.
NFR-4	Timezone correctness. All shift logic and timestamps shall use Asia/Kuwait consistently across server, web, and mobile.
NFR-5	Usability. Field screens shall present one primary action, large touch targets, and minimal text suitable for non-technical users.
NFR-6	Offline-tolerant assets. Fonts and assets shall be bundled locally so the UI renders on networks without external CDNs.
NFR-7	Extensibility. Notification delivery shall sit behind a provider interface so external channels integrate without changing activation, escalation, or broadcast logic.
NFR-8	Data integrity. Activation, escalation, and broadcast shall execute atomically; a running activation shall be immutable to plan edits.
NFR-9	Portability. Local development shall run with a single database container, with a zero-dependency database mode for evaluation.

11 · Security & Permissions Matrix

Authorization is enforced at two layers: per-endpoint role gates and per-record queryset filtering. A check mark means the action is permitted; data is additionally narrowed to the user's own scope where noted.

Capability	Admin	Manager	Leader	Member
Manage orgs/departments/users	✓	read-only	–	–
Create / edit plans	✓	✓	–	–
Activate a plan	✓	if authorized	–	–
View live dashboard	✓	✓	own teams	–
Run escalation / broadcast / close	✓	✓	–	–
Acknowledge readiness	–	–	✓	✓
Complete tasks / add notes	✓	✓	own team	own tasks
Reassign tasks	✓	✓	own team only	–
Set substitute (live)	✓	✓	own team	–
Raise issue to manager	–	–	✓	–

12 · Internationalization & Localization

- **Arabic-first.** Both applications default to Arabic with full right-to-left layout and field-staff-appropriate Arabic labels.
- **Complete English locale.** A full English translation ships alongside Arabic; a language toggle flips direction on web and switches instantly on mobile.
- **Single source of strings.** All user-facing text lives in one translation file per app; adding a language is one additional object with the same keys.
- **Regional conventions.** Kuwait timezone and the three fixed shift bands are built into the domain logic, not configured per install.

13 · Out of Scope (Deferred Roadmap)

The following are intentionally excluded from MVP 1.0 and recorded as the forward roadmap. Each lands on a seam the MVP deliberately left open and requires no re-architecture.

- Real notification channels (voice, SMS, WhatsApp) behind the existing provider interface.
- Automatic timed escalation via a scheduler (currently manual trigger).
- Automatic shift handover when an activation crosses a shift boundary.
- Surveys, lessons-learned, and the full post-activation final report (the MVP closure shows counts only).
- Checklist granularity below the task level.
- Permission granularity beyond the four roles.
- Push delivery (WebSockets / mobile push) replacing polling at scale.

14 · Acceptance Criteria & Verification

Each phase was accepted against concrete criteria, verified on the running system rather than by inspection.

Phase	Acceptance criterion (met)
Backend	Logging in as each role, activating the seeded plan, acknowledging as a member, running escalation, and retrieving the dashboard return correct numbers; member-only visibility holds; five attempts induct the substitute.
Web	A non-technical manager can create and activate a plan and watch the live dashboard, entirely in Arabic; production build is clean.
Mobile	A member acknowledges and completes tasks from the phone and the web dashboard reflects it within one polling interval; strict type-check is clean.
Polish	Broadcast reaches the member inbox; closing an activation produces the summary and stops polling; the language toggle works on both apps.

Residual verification note: physical-device layout (spacing and RTL edge cases on real phones) was not exercised in the build environment and warrants a polish pass on first device runs. All logic and API behavior were verified against the live backend.

15 · Glossary

Term	Meaning
Plan	A reusable template: teams, task templates, roster, metadata.
Activation	An immutable snapshot created when a plan is launched.
Participant	A person frozen into an activation from the roster.
Response	The single readiness tap «أنا جاهز» — the only counted acknowledgment.
Escalation	The process of calling non-responders and inducting substitutes after the attempt threshold.
Substitute	A pre-designated backup, resolved from the roster and inducted on escalation.
Shift	One of three fixed daily bands in Asia/Kuwait.
Broadcast	A manager message delivered to all participants of an activation.